

Fever with Rash 101:

Spot, Stop, Protect

Sipang Pangprasertkul, MD

Division of Infectious Diseases, Department of Pediatrics
Faculty of Medicine, Chiang Mai University



Outlines

- A syndromic approach: Fever with rash
- Fever with rash
 - Maculopapular: measles, rubella
 - Vesiculobullous: varicella

A Syndromic Approach: **Fever with Rash**



Initial assessment



Identify **"RED FLAGS"** → Rule out dangerous causes first!



Rash morphology

Benign vs **Life-threatening condition**

-  Unstable vital signs
-  Toxic appearance
-  Neck stiffness, seizure
-  Poor perfusion/shock
-  Non-blanching purpura

Stabilization

A Syndromic Approach: **Fever with Rash**

Why this approach?



Save time in
acute settings



Help prioritize
dangerous
conditions

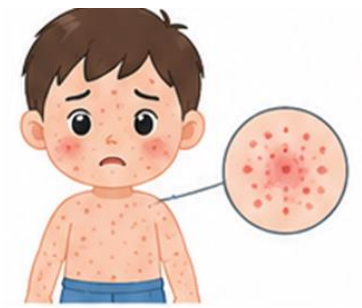


Reduces
unnecessary
investigations



Improves early
empirical
management

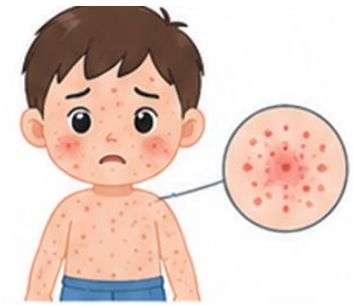
Approach to **Fever with Rash**



- **History taking**

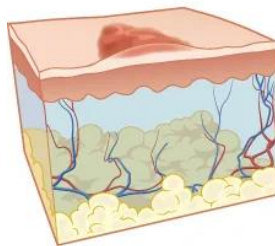
- Age
- Onset of symptoms (fever, rash)
- Characteristic of rash (morphology, distribution)
- Epidemiology, season
- Prodrome symptoms: cough, coryza, conjunctivitis
- Associated symptoms: respiratory, GI, neurological, mucosal involvement
- Exposure history (travel, animals, sick contact)
- Medications
- Immunization: MMR, Varicella
- Residential area
- Outbreak in community

Approach to **Fever with Rash**

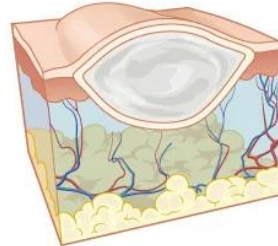


- **Physical examination:**

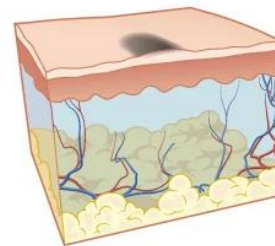
- **Characteristic of rash:** morphology, location, pattern, distribution, pruritus



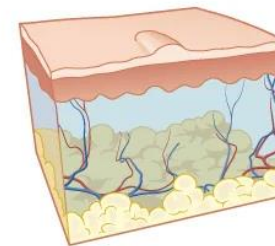
crust



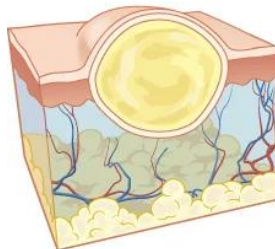
cyst



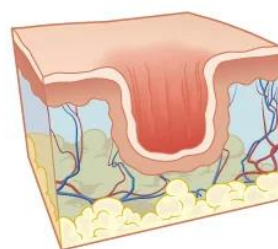
macule



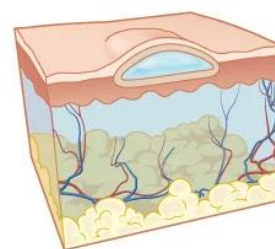
papule



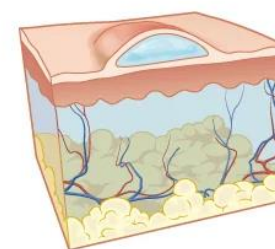
pustule



ulcer



vesicle

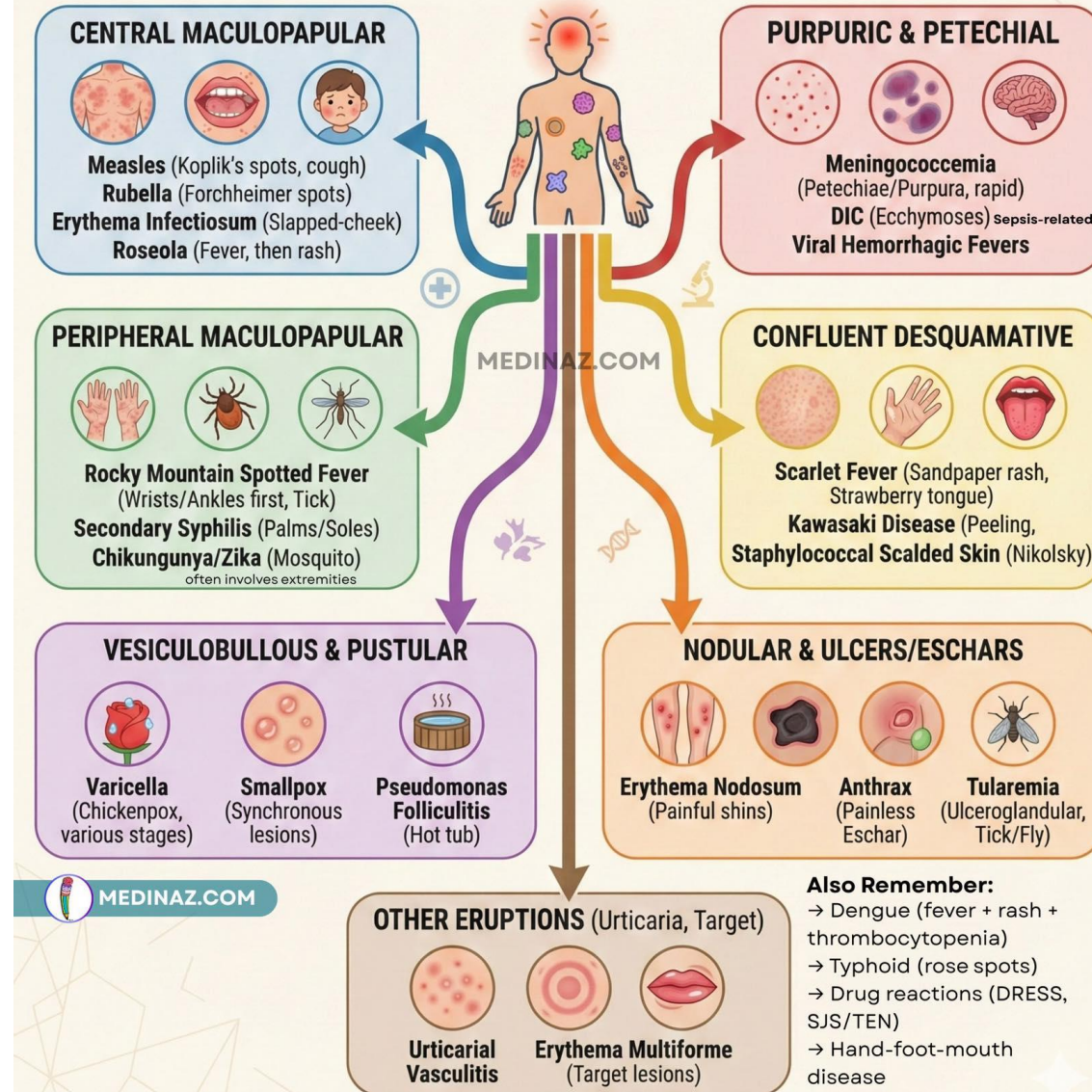


wheal

- Vital signs
- HEENT: pharynx, tonsils, conjunctivae, oral mucosa
- Lymph nodes
- Lungs
- CVS
- Abdomen: HSM
- Eschar

DISEASES ASSOCIATED WITH FEVER & RASH

A Visual Guide for Medical Students | Based on Clinical Presentations



Note: Presentations can vary. Consult a physician for diagnosis.



Infection

Virus

- Measles
- Rubella
- Roseola infantum (HHV-6)
- Erythema infectiosum (Parvovirus B19)
- Infectious mononucleosis (EBV)
- Adenovirus
- Vector-borne: dengue, zika, chikungunya
- Hand-foot-and-mouth disease (Coxsackievirus, EV71)
- Chickenpox (Varicella)
- Herpes simplex virus
- Herpes zoster virus, Mpox

Bacteria

- Scarlet fever (Group A streptococcus)
- Staphylococcal scalded skin syndrome (SSSS)
- Toxic shock syndrome
- Meningococemia
- Mycoplasma pneumoniae
- Rickettsial infection (scrub typhus, murine typhus)
- Secondary syphilis

Non-Infection

- Cutaneous drug eruption
- SLE, JIA
- Kawasaki disease
- MIS-C

Clues to think non-infectious

- ☒ Persistent fever despite treatment
- ☒ Poor response to antibiotics
- ☒ Other systemic features (joint pain, oral changes, edema, lymphadenopathy, etc.)



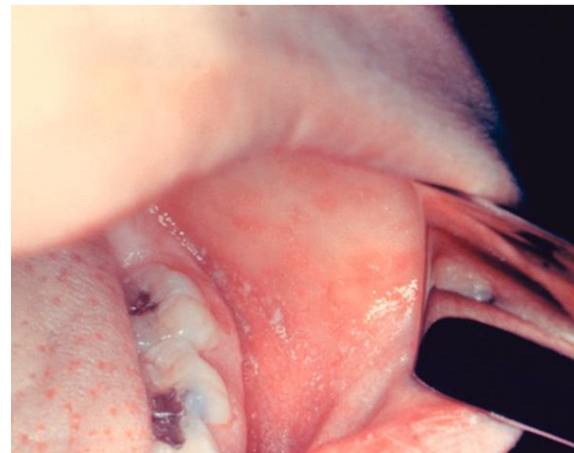
Measles



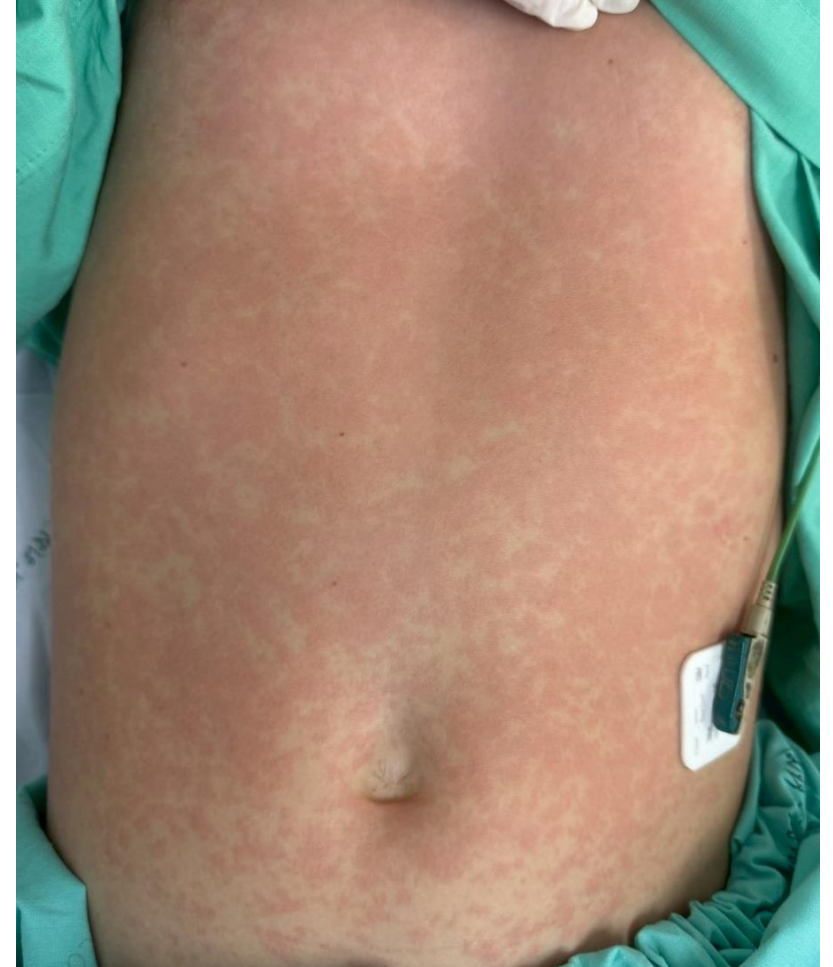
Red blotchy pattern on his face and body.



Conjunctivitis



Koplik spots on the buccal mucosa during the 3rd day of rash.



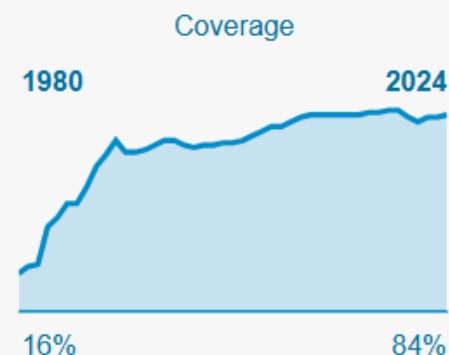
Confluent rash during the 2nd day of rash

The WHO/UNICEF Estimates of National Immunization Coverage (WUENIC) at the Global Level

**Measles-containing
vaccine, 1st dose**

84%

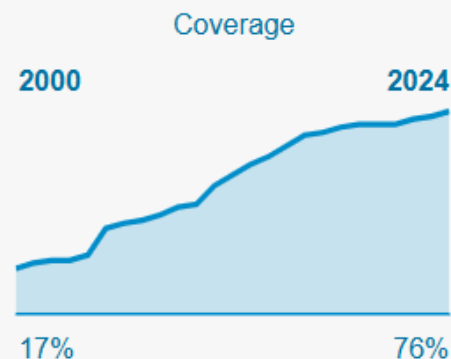
Thailand 1st dose (2024): 93%



**Measles-containing
vaccine, 2nd dose**

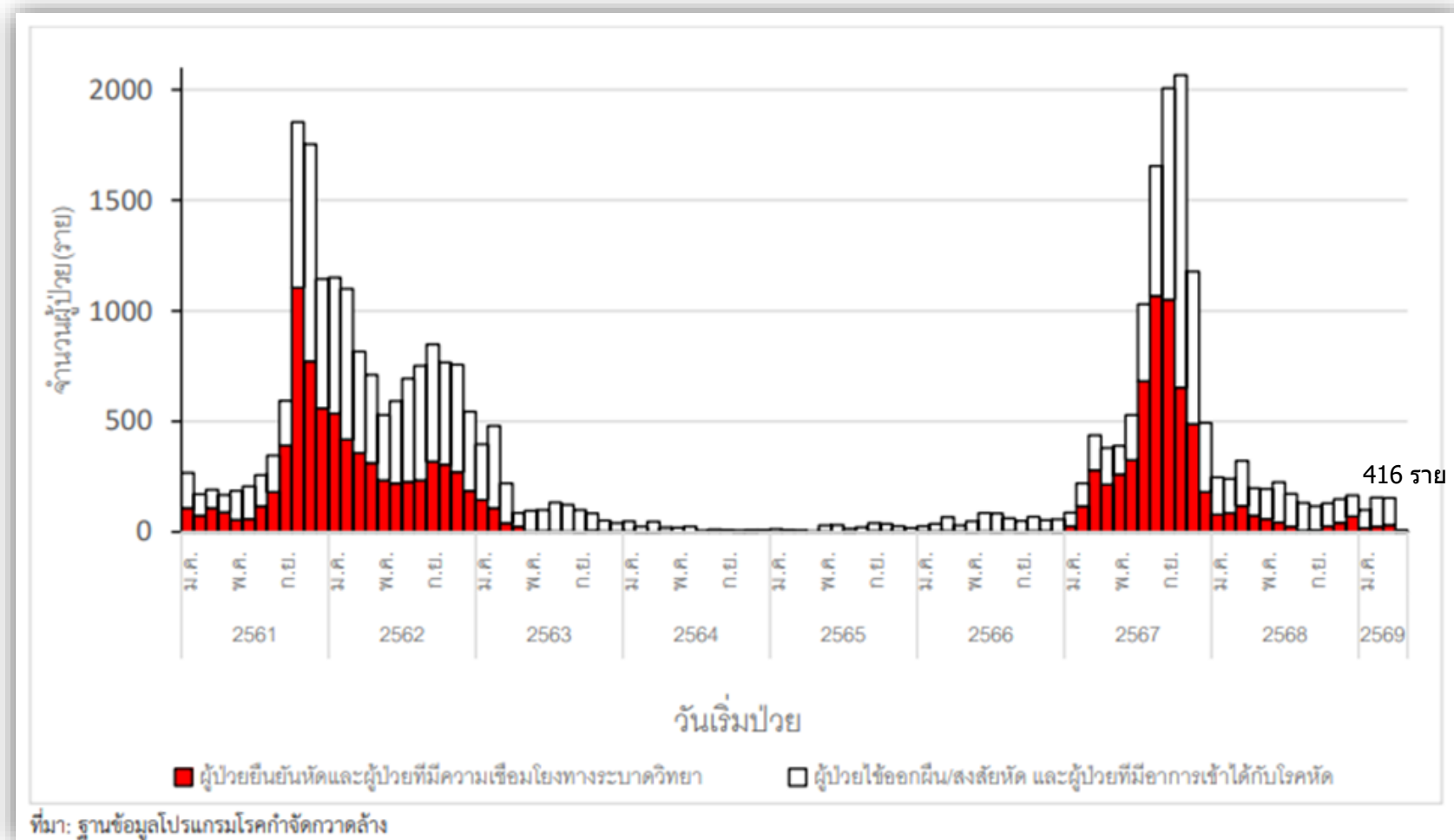
76%

Thailand 2nd dose (2024): 87%



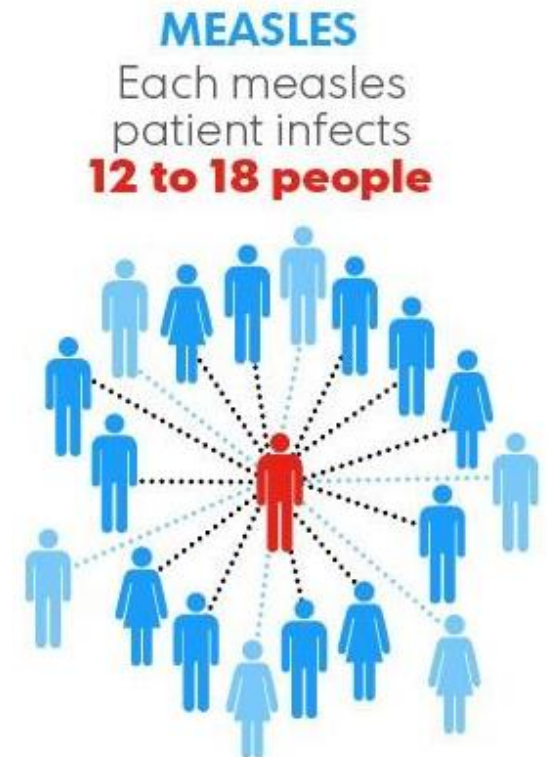
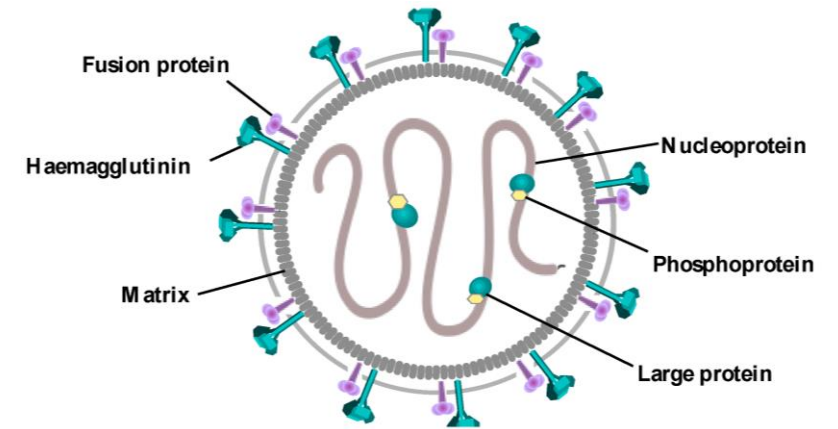
จำนวนผู้ป่วยไข่ออกผื่น/สงสัยหัด ในประเทศไทย

ตั้งแต่วันที่ 1 มกราคม 2561 – 7 เมษายน 2569



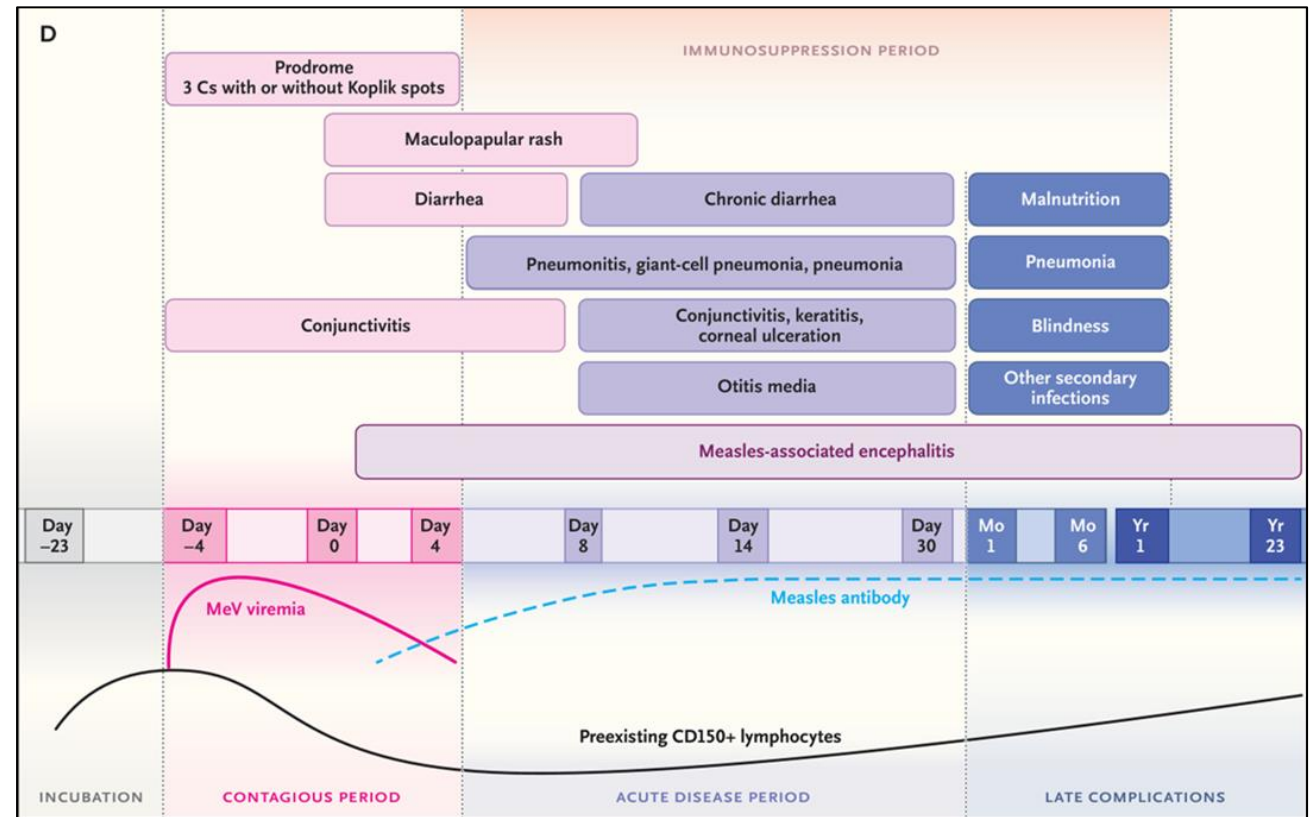
Measles

- **Highly contagious**
- **Etiology:** Measles virus (negative sense single-stranded, lipid-enveloped RNA virus), family Paramyxoviridae and genus *Morbillivirus*
- **Epidemiology:** Population levels of measles immunity of ~95% are required to interrupt the endemic spread of measles.
- **Transmission:** **airborne**



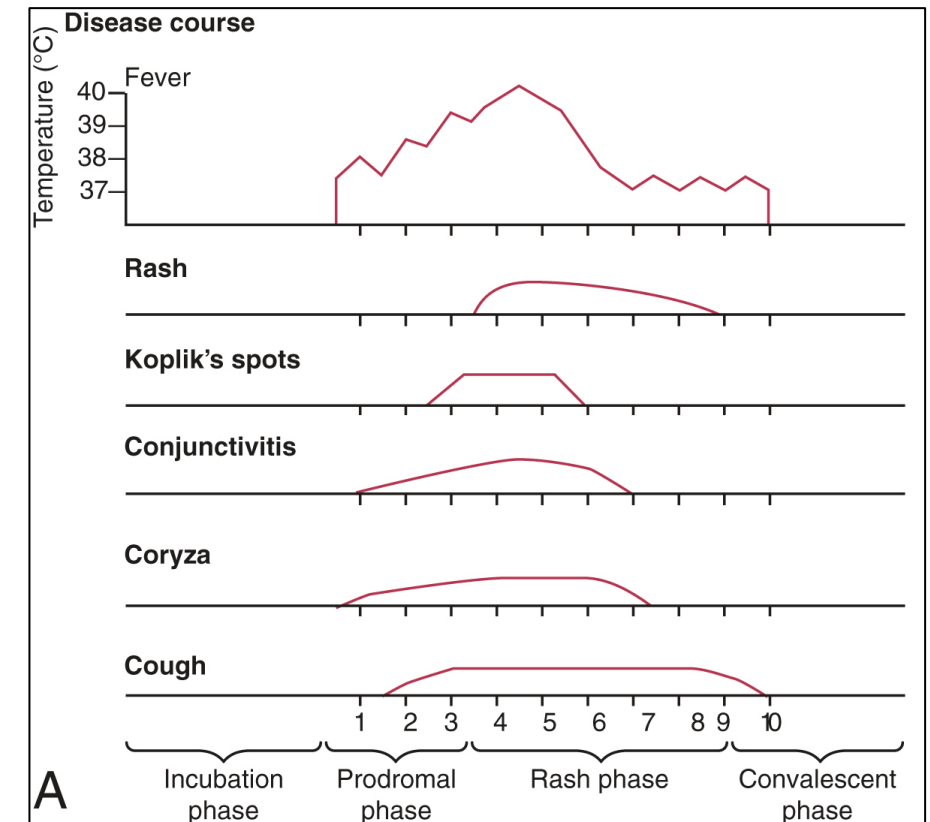
Measles: Pathogenesis

- Measles infection consists of 4 phases
 - 1) **Incubation period:** measles virus migrates to regional lymph nodes.
 - A primary viremia ensues that disseminates the virus to the reticuloendothelial system.
 - A secondary viremia spreads virus to body surfaces.
 - 2) **Prodromal illness**
 - 3) **Exanthematous phase**
 - 4) **Recovery**



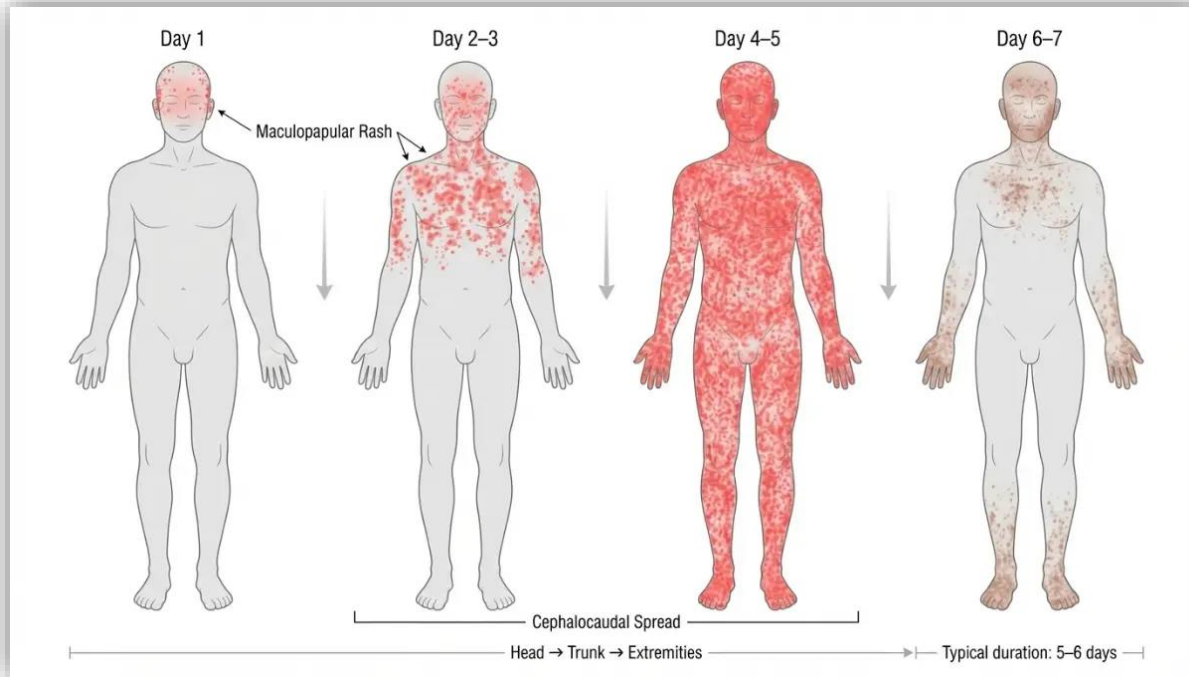
Measles: Clinical manifestations

- High fever, an enanthem, cough, coryza, conjunctivitis (3C), and a prominent exanthem
- **Prodromal phase:** mild fever, conjunctivitis with photophobia, coryza, a prominent cough, and increasing fever.
- **Koplik spots (50-70%):** pathognomonic sign of measles, appearing 1-4 days before the onset of the rash. They first appear as **discrete red lesions with bluish-white spots** in the center on the **inner aspects of the cheeks at the level of the premolars.**



8-12 days
(7-21 days)

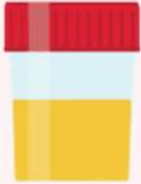
Measles



The diagnosis of measles is almost always **based on clinical and epidemiologic findings.**

- The rash **begins on the forehead** (around the hairline), behind the ears, and on the upper neck as a red maculopapular eruption.
- It then **spreads downward to the torso and extremities**, reaching the palms and soles (50% of cases).
- The exanthem frequently **becomes confluent** on the face and upper trunk
- The rash fades over about 7 days, hyperpigmentation → fine desquamation

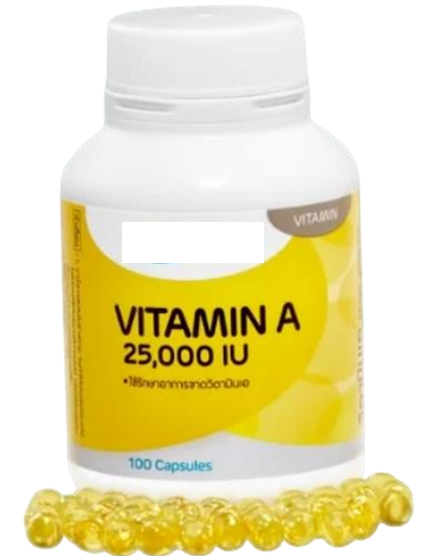
Measles: Diagnosis

Measles Tests			When to Collect?
Acute Disease	PCR	Nasopharyngeal (NP) or Throat (OP) Swab 	As soon as possible upon suspicion of measles: ideally 0-3 days after rash onset, up to 10 days after rash onset.
	PCR	Urine 	Within 10 days of rash onset <i>*Collecting a urine specimen along with an NP/OP swab may improve test sensitivity, especially if at the end of the PCR detection window.</i>
	IgM	Serum 	Collect with specimen for PCR. Can be negative up to 3 days after rash onset. IgM can be detected for 6-8 weeks after acute measles.
Immunity	IgG	Serum 	When assessing evidence of immunity, can be detected ~2 weeks after MMR vaccination

Up to 20% false negative result in the first 72 hours after rash onset

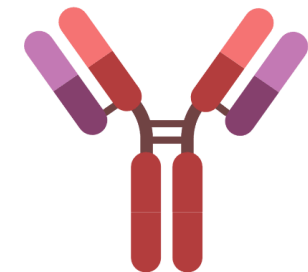
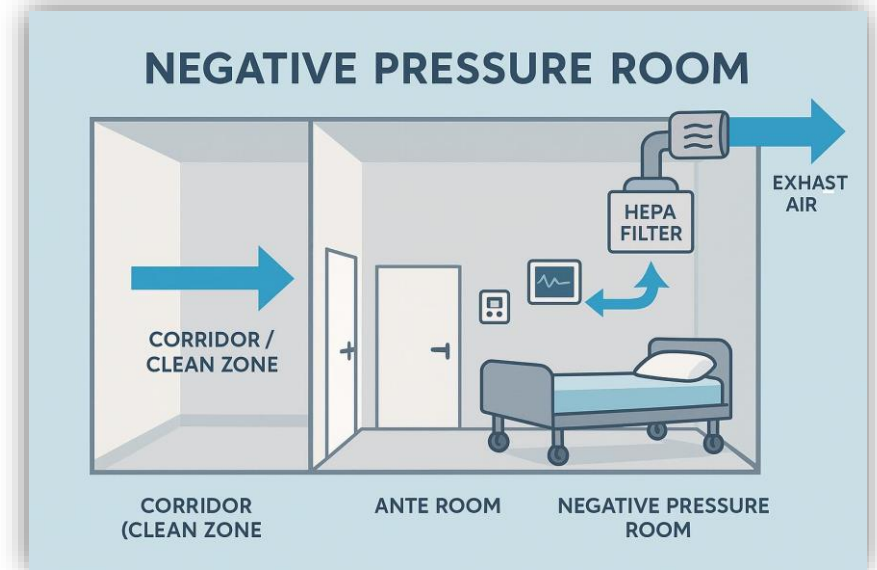
Measles: Treatment

- No specific antiviral therapy
- **Vitamin A:** The WHO currently recommend vitamin A for all children with measles, regardless of their country of residence.
 - Decreased morbidity and mortality rates
 - Vitamin A is administered **once daily for 2 days** (immediately on diagnosis and repeated the next day)
 - Dose
 - Age > 12 months: 200,000 IU
 - Age 6-11 months: 100,000 IU
 - Age < 6 months: 50,000 IU
 - An additional vitamin A should be given 2-6 weeks later to children with vitamin A deficiency.



Measles: Infection prevention and control measures

- **Airborne transmission precaution** for 4 days after the onset of rash in otherwise healthy children and for the duration of illness in immunocompromised patients.
- Exposed susceptible patient should be placed on airborne precautions for 21 days after last exposure (28 days if the patient received IVIG as PEP)
- Care of exposed people (**postexposure prophylaxis**)
 - Vaccine (within 72 hours)
 - Immunoglobulin; IG (within 6 days)



Measles: Complications

Table 1. Incidence of Severe Complications Associated with Measles.

Complications	Incidence in Developed Countries	Comments
Pneumonia	1–6 per 100 measles cases ⁵	Among the most common complications during the first month of measles; most common cause of measles hospitalization
Diarrhea	8–10 per 100 measles cases ⁵	Common complication during the first month of measles
Keratitis or keratoconjunctivitis	3–10 per 100 measles cases ⁶⁻⁹	Keratoconjunctivitis may appear in the prodromal stages of measles and persist for as long as 3 months ⁶ ; keratitis with retinitis and optic neuritis also has been reported. ⁷⁻⁹
Corneal ulceration	Rare	Documented in 1–4 per 100 measles cases in the 1980s in Africa and South Asia ^{10,11} ; measles can cause corneal ulceration directly and facilitate a secondary infection (such as herpes simplex keratitis) that leads to corneal ulceration ^{10,11}
Blindness	Rare	Measles is a leading cause of childhood blindness in places where measles is endemic; results of surveys conducted in schools in Africa in the 1970s suggested that measles was the cause of 33 to 79% cases of blindness. ¹¹
Otitis media	7–9 per 100 measles cases ⁵	One of the most common complications during the first month of measles; can lead to sensorineural deafness, which was observed in 5 to 10% of measles cases in the United States before the introduction of measles vaccination programs ⁵
Death	1–3 per 1000 measles cases ⁵	16 per 1000 measles cases in low-income countries ¹² ; 9 per 1000 measles cases in middle-income countries ¹² ; up to 180 per 1000 measles cases reported in the context of humanitarian relief efforts during major outbreaks ¹³
Malnutrition	8–10 per 100 measles cases	
Acute postinfectious measles encephalitis	1 per 1000 measles cases ¹⁴	Develops within the first week of measles, after the appearance of the first symptoms, and is associated with 20% mortality ¹⁴
Measles-inclusion body encephalitis	1 per 1000 measles cases ¹⁴	Develops within 7 days to 6 months after onset of measles and is associated with 100% mortality ¹⁴
Subacute sclerosing pan-encephalitis	7–11 per 100,000 measles cases ^{5,14}	Develops within 7–10 years after measles and is associated with 100% mortality within 1–3 years after onset ¹⁴ ; young children with measles (<2 years of age) are at increased risk

Complications

Neurological:
ADEM, MIBE,

SSPE

Otitis media

Pneumonia

Adverse
pregnancy
outcomes

Keratoconjunctivitis
(blindness)

Stomatitis

Laryngitis (croup)

Diarrhea

Death

B

Measles: Postexposure Prophylaxis (PEP)

Age Range	Measles Immune Status ^a	PEP Type Depending on Time After Initial Exposure		
		≤3 days (≤72 hours)	4–6 days	>6 days
All ages (≥6 mo)	Immune	• PEP not indicated. Exposed person has documented immunity.		
<6 mo	Nonimmune (because of age ^b)	• Administer immune globulin intramuscular (IGIM) ^c • Home quarantine ^d		• PEP not indicated (too late). • Home quarantine ^d
6–11 mo	Nonimmune	• Administer MMR vaccine (MMR vaccine preferred over immune globulin [IG]) • No quarantine needed. ^e	• Administer IGIM ^c • Home quarantine ^d	• PEP not indicated (too late). • Home quarantine ^d
≥12 mo	Nonimmune	• Administer MMR vaccine • No quarantine needed ^e	• IG PEP usually not administered ^f • Home quarantine, ^d then administer MMR vaccine to protect from future exposures	
≥12 mo	1 dose of MMR vaccine	• Administer 2 nd MMR vaccine dose if ≥28 days from the first dose • No quarantine needed (person had 1 dose when exposed)		

Measles: Postexposure Prophylaxis (PEP)

Category	Measles Immune Status ^a	PEP Type Depending on Time After Initial Exposure		
		≤3 days (≤72 hours)	4–6 days	>6 days
Severely immunocompromised ^b	IG recommended regardless of measles immune status	- Administer immune globulin intravenous (IGIV)^c - Home quarantine^d		- PEP not indicated (too late) - Home quarantine^d
Pregnant	Immune	PEP not indicated		
	Nonimmune	- Administer IGIV^c - Home quarantine^d		- PEP not indicated (too late) - Home quarantine^d

- **Severely immunocompromised patients:** GVHD, on treatment for ALL within and until least 6 months after completion CMT, HIV with severe immunosuppression
- **IVIG:** 400 mg/kg

 ตารางการสร้างเสริมภูมิคุ้มกันโรคในเด็กและวัยรุ่นไทย แนะนำโดย สมาคมโรคติดเชื้อในเด็กแห่งประเทศไทย **2569**

วัคซีนจำเป็นที่ต้องให้กับเด็กทุกคน

วัคซีน	อายุ	แรกเกิด	1 เดือน	2 เดือน	4 เดือน	6 เดือน	9-12 เดือน	18 เดือน	2-2½ ปี	4-6 ปี	11-12 ปี
บีซีจี ¹ (BCG)		BCG									
ตับอักเสบบี ² (HB)		HB1	(HB2)								
คอตีบ-บาดทะยัก-ไอกรนชนิดทั้งเซลล์ ³ (DTwP)				DTwP-HB-Hib-1	DTwP-HB-Hib-2	DTwP-HB-Hib-3		DTwP กระตุ้น 1		DTwP กระตุ้น 2	Td และ ทุก 10 ปี
ฮิบ ⁴ (Hib)											
โปลิโอ ⁵ ชนิดฉีด (IPV) และกิน (OPV)				IPV1	IPV2	OPV1		OPV กระตุ้น 1		OPV กระตุ้น 2	
โรต้า ⁶ (Rota)				Rota1	Rota2	(Rota3)					
หัด-คางทูม-หัดเยอรมัน ⁷ (MMR)							MMR1	MMR2			
ใช้สมองอักเสบเจอีชนิดเชื้อมีชีวิต ⁸ (live JE)							JE1		JE2		
ไข้หวัดใหญ่ ⁹ (Influenza)							Influenza ให้ปีละครั้ง (การให้ครั้งแรกในเด็กอายุต่ำกว่า 9 ปี ให้ 2 เข็ม ห่างกัน 1 เดือน)				
เอชพีวี ¹⁰ (HPV)											เด็กหญิงประมณ 5 ตามแผนฯ ของ กระทรวงสาธารณสุข

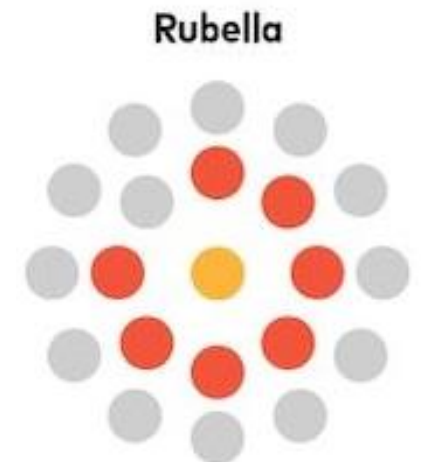
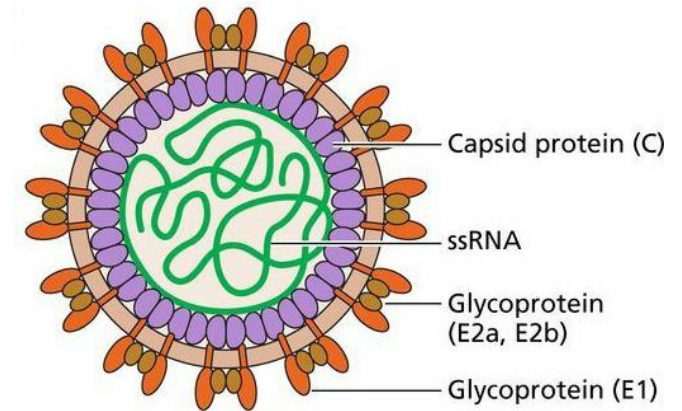
7. วัคซีนหัด-คางทูม-หัดเยอรมัน

- ให้วัคซีนครั้งแรกเมื่ออายุ 9 -12 เดือน และครั้งที่ 2 เมื่ออายุ 18 เดือน ในพื้นที่ที่มีรายงานโรคหัดจำนวนน้อยอาจฉีดเข็มแรกเมื่ออายุ 12 เดือน
- ในกรณีที่มีการระบาดหรือสัมผัสโรค ควรเริ่มฉีดวัคซีนและฉีดให้ครบโดยเร็ว ดังนี้
 - สามารถเริ่มฉีดเข็มแรก ตั้งแต่อายุ 6 เดือนถึงน้อยกว่า 9 เดือน ให้ฉีดซ้ำเข็มที่ 2 ที่อายุ 12 เดือน และเข็มที่ 3 ที่อายุ 18 เดือน
 - ถ้าเริ่มฉีดเข็มแรกตั้งแต่อายุ 9-12 เดือน ให้ฉีดเข็มที่ 2 ห่างจากเข็มแรกอย่างน้อย 3 เดือน ได้ 2 เข็มถือว่าฉีดครบ
 - ถ้าเริ่มฉีดเข็มแรกตั้งแต่อายุ 12 เดือน ให้ฉีดเข็มที่ 2 ห่างจากเข็มแรกอย่างน้อย 1 เดือน ได้ 2 เข็มถือว่าฉีดครบ

- ในกรณีที่ต้องการควบคุมการระบาดของหัด อาจใช้วัคซีนหัด-คางทูม-หัดเยอรมัน (MR) แทนได้ในเด็กที่อายุมากกว่า 4 ปีและผู้ใหญ่
- ในกรณีที่ต้องการฉีดวัคซีน หัด-คางทูม-หัดเยอรมัน และอีสุกอีใสในเวลาเดียวกัน สามารถใช้วัคซีนรวม หัด-คางทูม-หัดเยอรมัน-อีสุกอีใส (MMRV) แทนการฉีดแบบแยกเข็มได้ทุกวัยในเด็กอายุตั้งแต่ 1-12 ปี การใช้วัคซีนรวม MMRV เข็มแรกในเด็กอายุ 12- 47 เดือนมีโอกาสเกิดการชักจากไขได้มากกว่าการฉีดแยกเข็ม แต่ในเข็มที่สองในเด็กอายุ 15 เดือน-12 ปี หรือเข็มแรกที่ยาวตั้งแต่ 4 ปีขึ้นไป พบมีอาการข้างเคียงไม่แตกต่างกัน
- กรณีที่เคยได้วัคซีน MMR หรือ VZV มาก่อน แนะนำให้วัคซีนรวม MMRV ห่างจากวัคซีน MMR และ VZV ครั้งก่อน อย่างน้อย 3 เดือน

Rubella (German measles)

- Mild, often exanthematous disease of infants and children
- Major clinical significance is transplacental infection and fetal damage: congenital rubella syndrome (CRS)
- **Etiology:** Rubella virus (enveloped, positive single-stranded RNA virus), family Togaviridae and genus *Rubivirus*
- **Transmission:** postnatal rubella → droplet



Rubella: Clinical manifestations

- Many case of postnatal rubella are subclinical.
- **Generalized erythematous maculopapular rash**, lymphadenopathy, slight fever
- The rash starts on the face, becomes generalized in 24 hours, and last a median of 3 days
- Lymphadenopathy involves **posterior auricular** or **suboccipital lymph nodes**
- Transient polyarthralgia and polyarthrititis (adolescents and adults)



Rubella

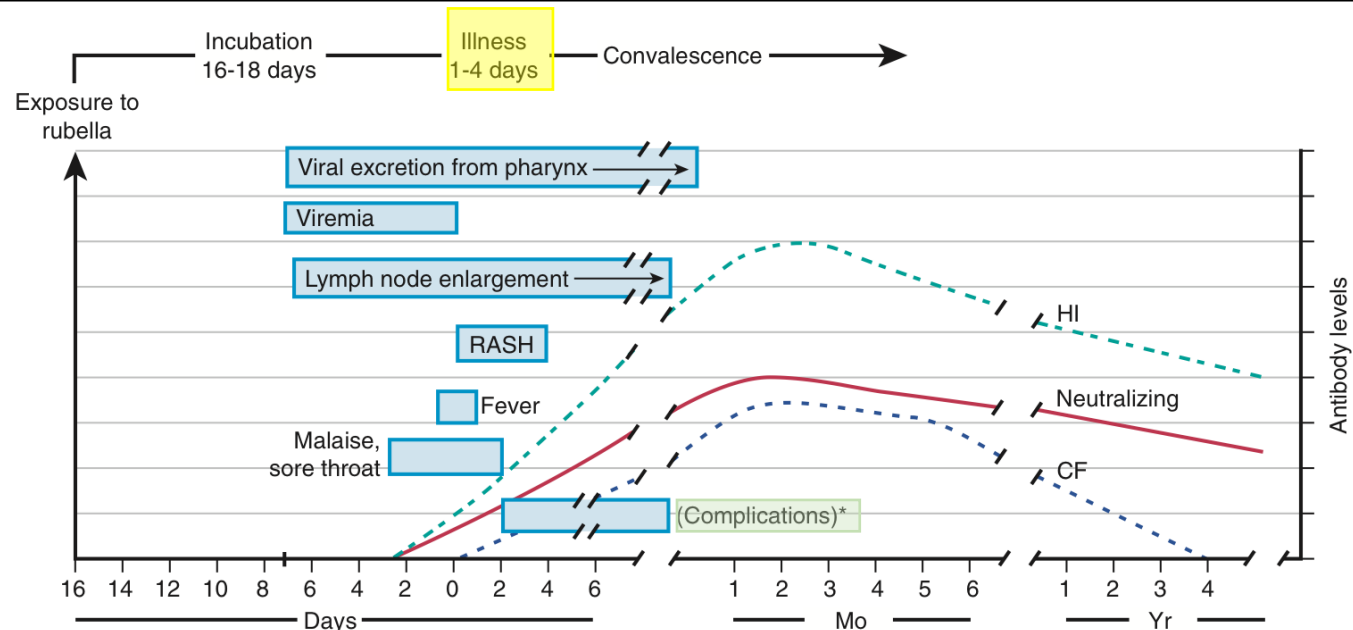


Figure 247-2 Pathophysiologic events in postnatally acquired rubella virus infection. *Possible complications include arthralgia and/or arthritis, thrombocytopenic purpura, and encephalitis. CF, complement fixation titer; HI, hemagglutination-inhibition titer. (From Lamprecht CL: Rubella virus. In Beshe RB, editor: Textbook of human virology, ed 2, Littleton, MA, 1990, PSG Publishing, p. 685.)

Diagnosis:

1. Rubella viral RNA by RT-PCR
2. Rubella IgM (most postnatal rubella cases are IgM-positive by 5 days after symptom onset)
3. Rubella IgG antibody titer (a 4-fold increase in paired serum, at least 10 days apart)
4. Rubella virus in cell culture

Rubella

- **Treatment:** supportive
- **Prognosis:** excellent prognosis in postnatal rubella infection
- **Prevention:** postnatal rubella droplet precautions are recommended for 7 days after onset of the rash.
- **Vaccination:** MMR

Varicella



Vesicular lesions with erythematous margins

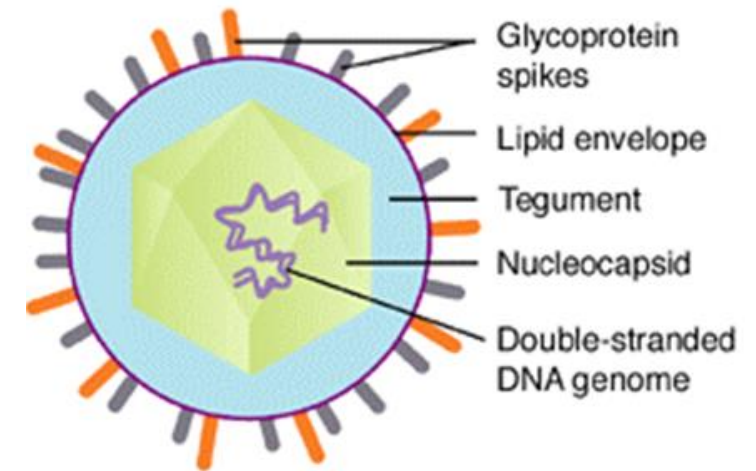
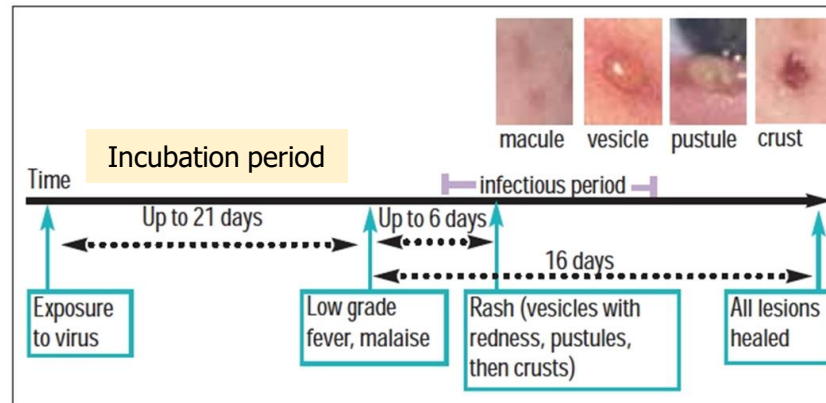


Rash later in course with many lesions crusting

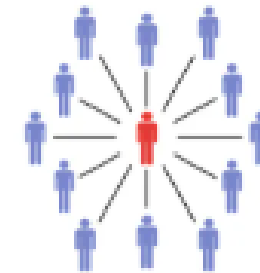


Varicella

- **Highly contagious**
- **Etiology:** VZV (HHV-3), *Herpesviridae* family, subfamily *Alphaherpesvirinae*, genus *Varicellovirus*
- **Transmission:** **airborne transmission**
- **Incubation period:** 14-16 days (range 10-21 days after exposure to rash)



12



Chickenpox

Varicella: Clinical manifestations

- Primary infection: varicella (chickenpox)
- Unvaccinated people:
 - generalized, pruritic, erythematous **vesicular** rash (250-500 lesions) in **varying stages** of development (**papules, vesicles**) and resolution (**crusting**)
 - Low-grade fever, other systemic symptoms
- Varicella trends to be more severe in adults, infants, and adolescents
- Breakthrough cases can occur in immunized children: mild, rash presentation is modified
- Immunocompromised children: progressive, severe varicella, hemorrhagic skin lesions, high fever, visceral dissemination (encephalitis, hepatitis, pneumonia)

Varicella: Clinical manifestations

Chicken pox

Varicella

Pathogen

Varicella-zoster virus

Course

Widespread vesicular rash with lesions beginning as macules and rapidly becoming papules
Simultaneous occurrence of various rash stages
Remission of exanthem after 8 days

Complications

Bacterial superinfection
Meningitis, acute cerebellar ataxia, encephalitis
Herpes zoster (Shingles) from virus reactivation
Congenital varicella syndrome

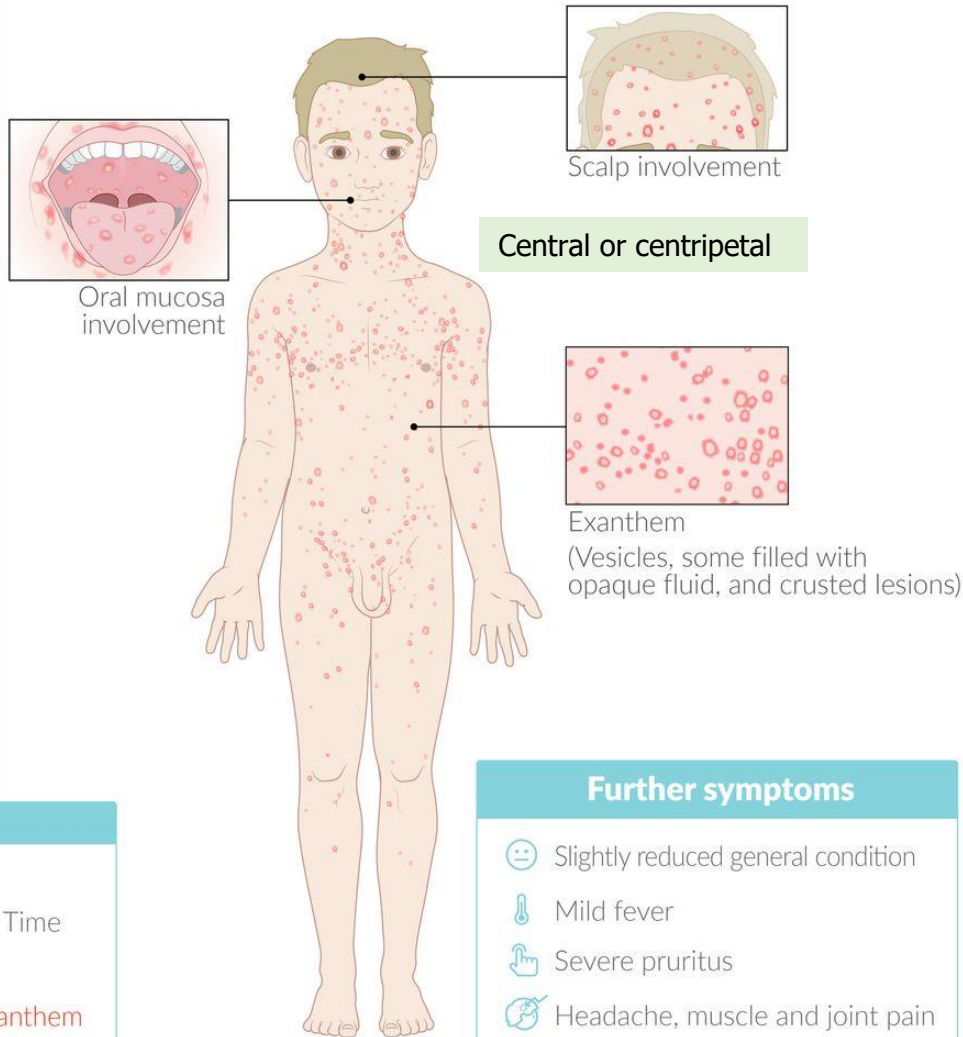
Treatment

Symptomatic

Vaccine

Yes

Course of disease

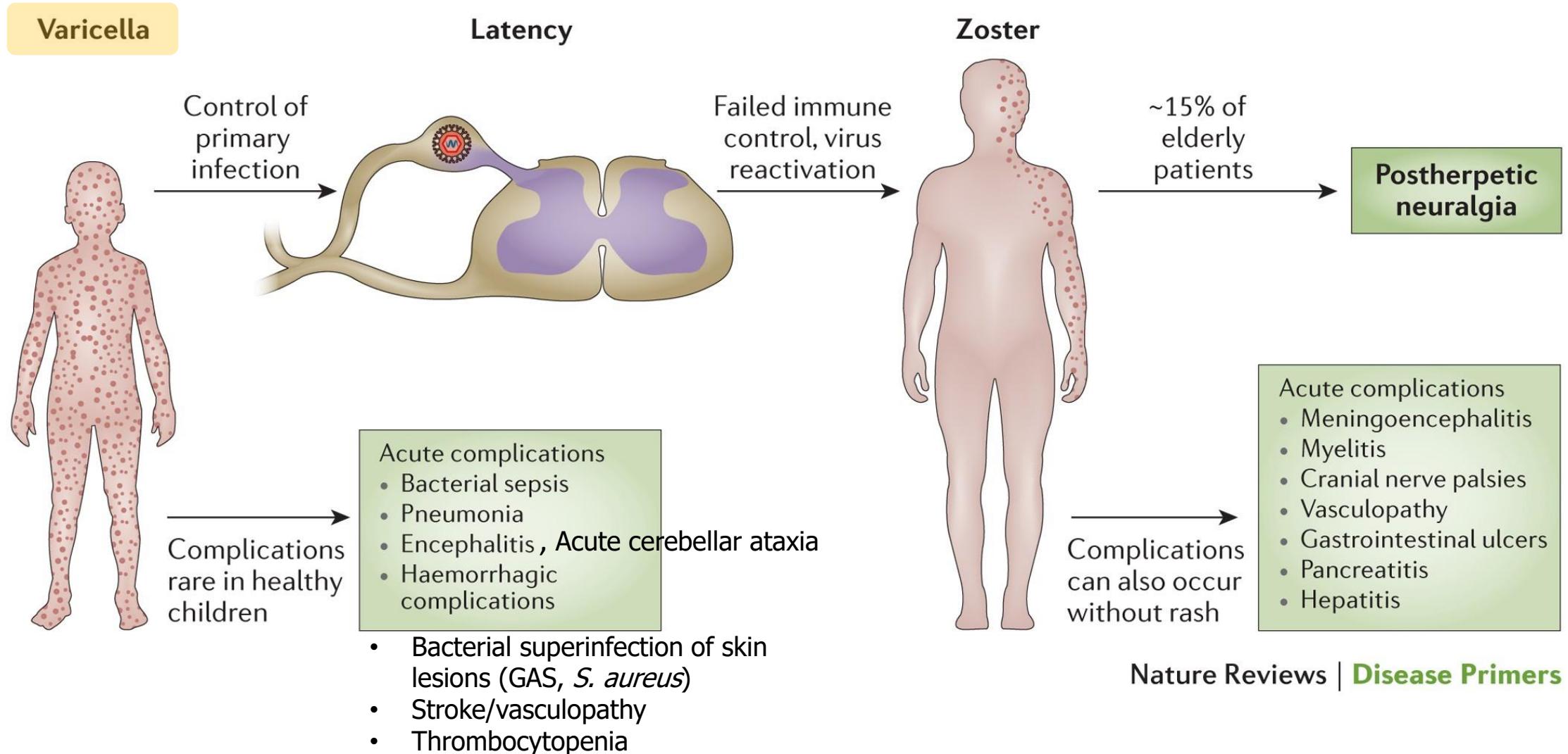


Further symptoms

- Slightly reduced general condition
- Mild fever
- Severe pruritus
- Headache, muscle and joint pain

- Varicella lesion often appear first on the scalp, face or trunk.
- Intensely pruritic erythematous macules
- Papular stage to form clear, fluid-filled vesicles
- Clouding and umbilication of the lesions begin in 24-48 hr
- Crusting

Varicella: Complications

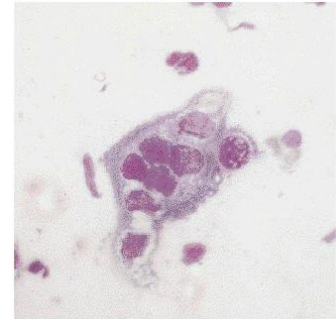


Varicella: Diagnosis

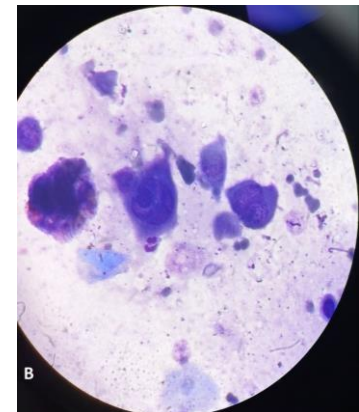
Test	Specimen	Comments
NAAT	Vesicular swabs or scrapings, scabs from crusted lesions, biopsy tissue, CSF	Very sensitive method. Specific for VZV. Methods have been designed that distinguish vaccine strain from wild-type (see text).
DFA	Vesicle scraping, swab of lesion base (must include cells)	Specific for VZV. More rapid and more sensitive than culture, less sensitive than NAAT.
Viral culture	Vesicular fluid, CSF, biopsy tissue	Distinguishes VZV from HSV. Limited availability, requires up to a week for result. Least sensitive.
Serology (IgG) ^a	Acute and convalescent serum specimens for IgG	Specific for VZV. Commercial assays generally have low sensitivity to reliably detect vaccine-induced immunity.
Serology (IgM)	Acute serum specimens for IgM	IgM serology is less specific than IgG serology. IgM inconsistently detected. Not reliable method for routine confirmation.

CSF indicates cerebrospinal fluid; DFA, direct fluorescent antibody; HSV, herpes simplex virus; IgG, immunoglobulin G; IgM, immunoglobulin M; NAAT, nucleic acid amplification test.

Tzanck Smear



Multinucleated giant cell, an epithelial cell containing numerous nuclei



Varicella: Treatment



Nonspecific Therapies



Keeping fingernails short
to prevent trauma and
secondary bacterial
infection from scratching



Frequent bathing



Application lotion
to reduce pruritus



Acetaminophen for fever



Avoid aspirin (increase
the risk of Reye syndrome)



Varicella: Treatment

- **Antiviral therapy** and the **route** and **duration** of therapy should be determined by **host factors** and **extent of infection**
- Immunocompetent host: most virus replication has stopped by 72 hours after onset of rash, may be extended in immunocompromised host
- **Oral acyclovir and valacyclovir** are not routinely recommended for otherwise healthy younger children with varicella
- Antiviral therapy should be considered for people **at increased risk of moderate to severe varicella**:
 - Unvaccinated people >12 years
 - Chronic cutaneous or pulmonary disorders
 - On long-term salicylate therapy
 - On short or intermittent courses of corticosteroids

Varicella: Treatment

- **IV acyclovir**
 - Immunocompromised patients
 - High-dose corticosteroid therapy >14 days
- Therapy initiated early in the course of the illness, especially within 24 hours of rash onset, maximizes benefit
- Oral acyclovir should not be used to treat immunocompromised children with varicella because of poor oral bioavailability

Management of Exposures to Varicella-Zoster Virus

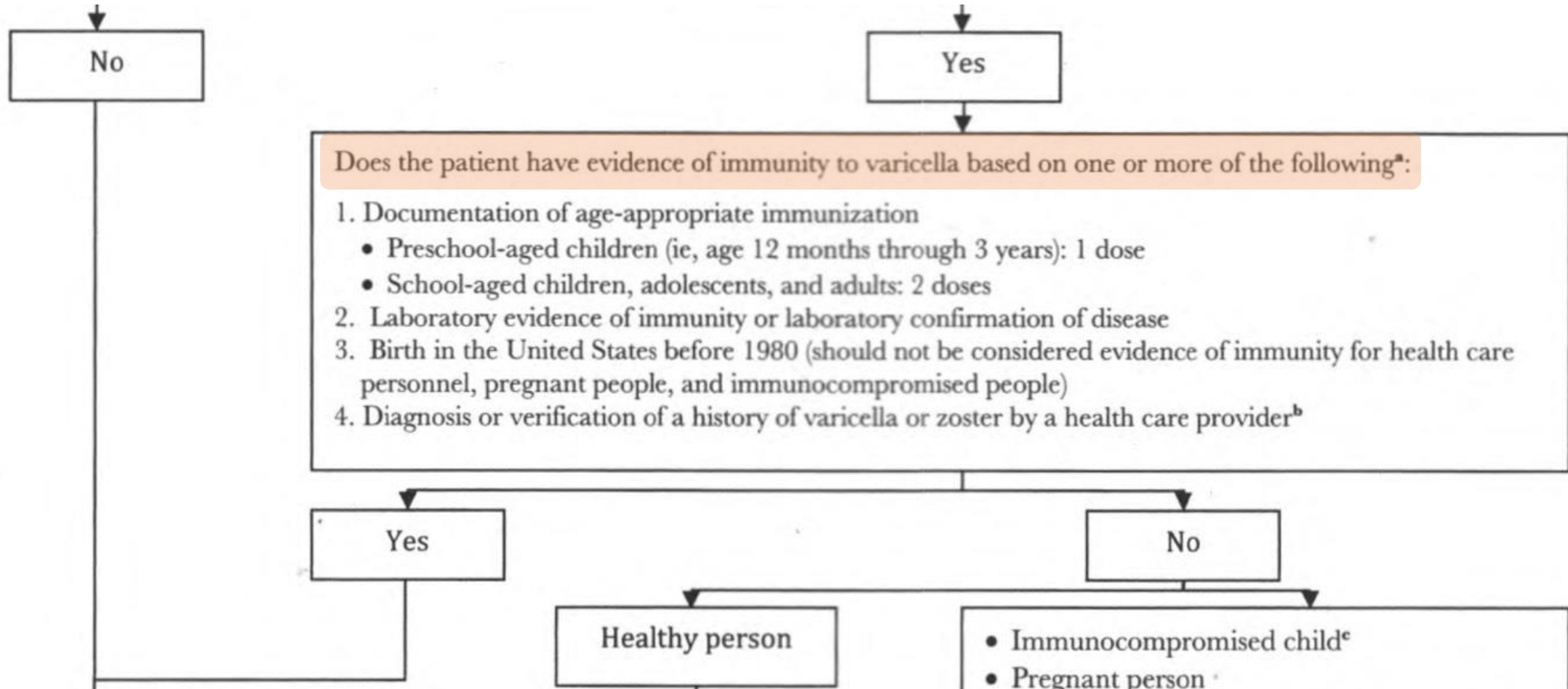
Significant exposure:

- Household: residing in the same household
- Playmate: face-to-face indoor play ≥ 5 minutes (some experts use > 1 hour)
- Newborn infant
- Hospital:
 - Varicella: In same 2- to 4-bed room or adjacent beds in a large ward, face-to-face contact with an infectious staff member or patient, or visit by a person deemed contagious
 - Zoster: Contact (eg, touching or hugging) with a person with disseminated zoster or with uncovered uncrusted lesions

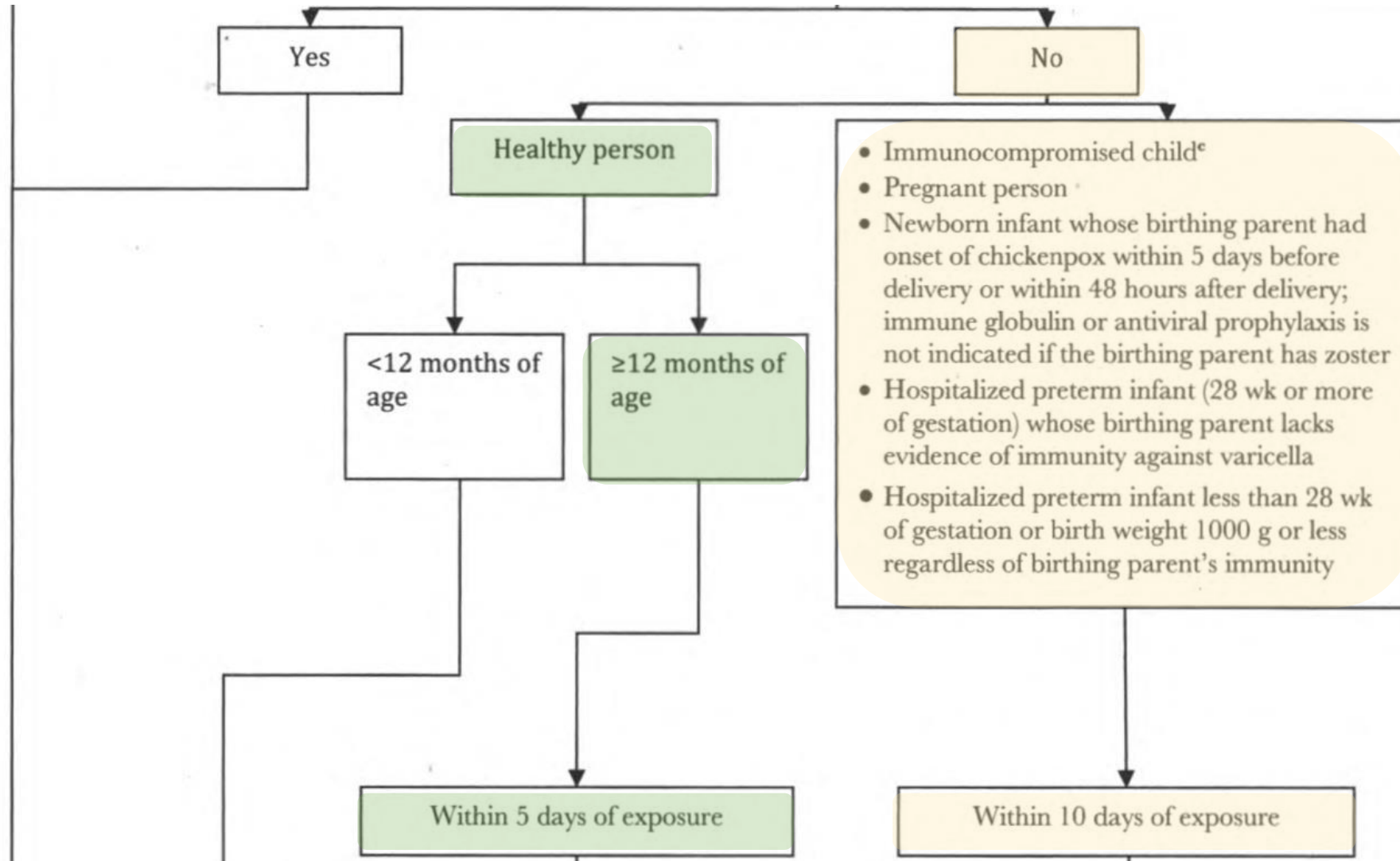
No

Yes

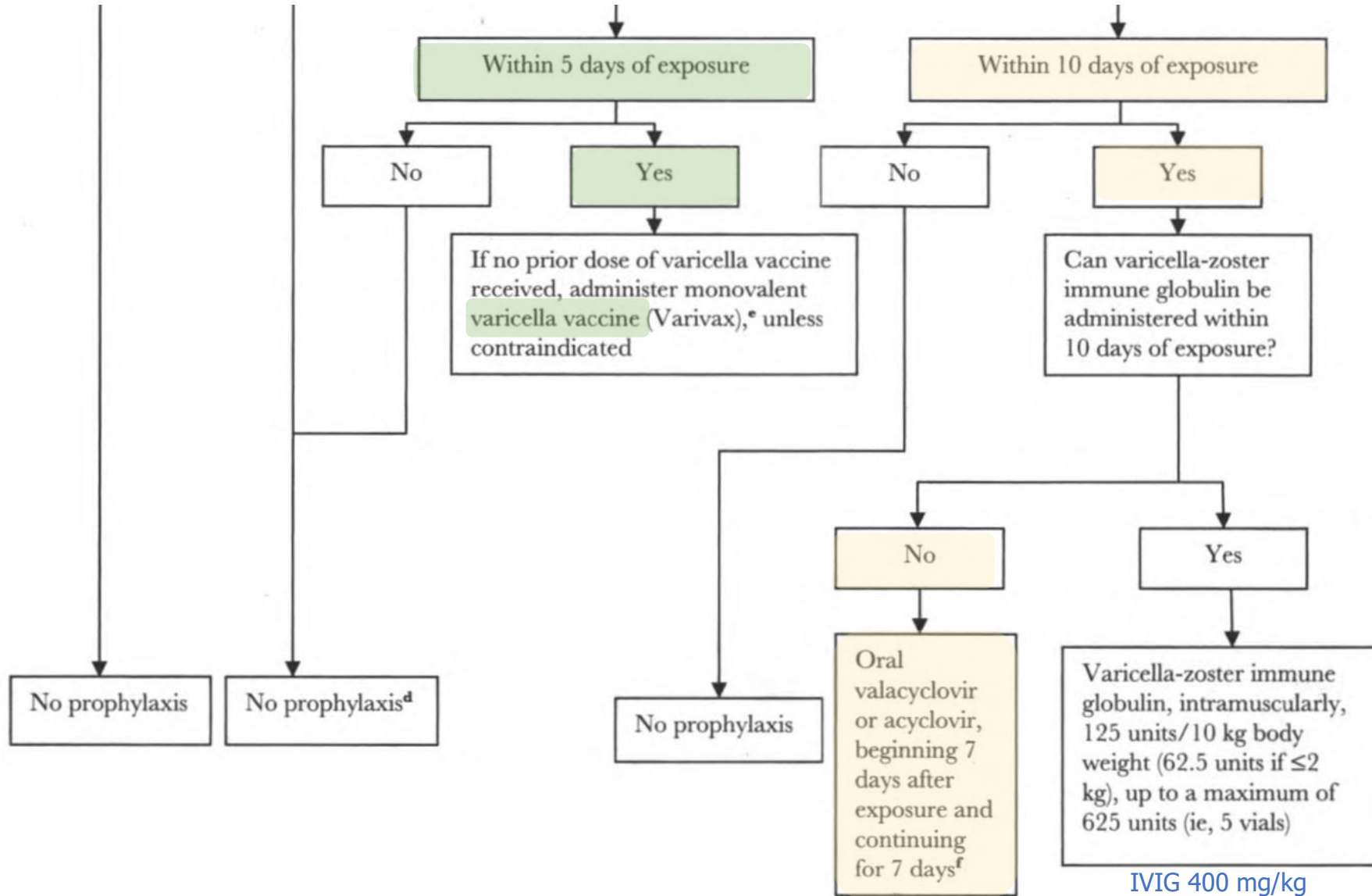
Management of Exposures to Varicella-Zoster Virus



Management of Exposures to Varicella-Zoster Virus



Management of Exposures to Varicella-Zoster Virus



Varicella: Infection prevention and control measures

- **Child care and school:**

- Children with uncomplicated varicella who have been excluded from school/child care may return when the rash has crusted or
- Immunized people without crusts, until no new lesions appear within a 24-hour period

- **Hospitalized patients:**

- In addition to **standard precautions, airborne and contact precautions** until all lesions are dry and crusted, typically at least 5 days after onset of rash

Varicella Vaccine

วัคซีนอื่น ๆ หรือภูมิคุ้มกันสำเร็จรูป ที่อาจให้เสริม หรือทดแทน													
วัคซีน	อายุ	แรกเกิด	2 เดือน	4 เดือน	6 เดือน	12-15 เดือน	18 เดือน	2-2½ ปี	4 ปี	6 ปี	9 ปี	11-12 ปี	18 ปี
คอตีบ-บาดทะยัก-ไอกรน ชนิดไร้เซลล์ ³ หรือ ชนิดทั้งเซลล์ (DTaP/DTwP, Tdap, หรือ Tdap) ตัวย่อเลขบี โปลิโอ ⁵ ชนิดฉีด (IPV) ฮิบ ⁴ (Hib)			DTaP/DTwP-HB-IPV-Hib1	DTaP/DTwP-(HB)-IPV-Hib2	DTaP/DTwP-HB-IPV-Hib3		DTaP-IPV-(Hib4) กระตุ้น 1		DTaP-IPV หรือ Tdap-IPV หรือ Tdap กระตุ้น 2			Tdap หรือ Tdap ต่อไป Td หรือ Tdap/Tdap ทุก 10 ปี	
นิวโมคอคคัสชนิดคอนจูเกต ¹¹ (PCV)			PCV1	PCV2	(PCV3)	PCV4							
ไข้มองอักเสบนกเชื้อไม่มีชีวิต ⁹ (Inactivated JE)					JE1, JE2 ห่างกัน 4 สัปดาห์ และ JE3 อีก 1 ปี								
อีวี 71 ¹² (EV71)					EV71 2 เข็ม ห่างกัน 1 เดือน								
ตับอักเสบบี ¹³ (HAV)						HAV ชนิดเชื้อไม่มีชีวิต ให้ 2 ครั้ง ห่างกัน 6-12 เดือน							
อีสุกอีใส ¹⁴ (VZV) หรือวัคซีนรวม หัด-คางทูม-หัดเยอรมัน-อีสุกอีใส (MMRV)						VZV1 (หรือ MMRV1)	VZV2 (หรือ MMRV2)						

14. วัคซีนอีสุกอีใส

- 1) วัคซีนเชื้อมีชีวิต มี 2 แบบ คือ สายพันธุ์ OKA (Varilrix™, Varivax™, SKYvaricella™) และ MAV06 (Varicella GCC) ฉีดได้ตั้งแต่อายุ 1 ปีขึ้นไป แนะนำให้ฉีดเข็มแรกอายุ 12-18 เดือน
- 2) อาจพิจารณาให้ฉีดเข็มที่ 2 เมื่ออายุ 18 เดือน-4 ปี อาจฉีดเข็มที่ 2 ก่อนอายุ 4 ปีได้ในกรณีที่มีการระบาด โดยต้องห่างจากเข็มแรกอย่างน้อย 3 เดือน อาจใช้วัคซีน MMRV แทน MMR และ VZV แบบแยกเข็ม (ดูในข้อ 7)
- 3) ถ้าอายุมากกว่า 13 ปี ให้ฉีดสองเข็มห่างกันอย่างน้อย 1 เดือน



Thank you

E-mail: sipang.p@cmu.ac.th

